

For lower resistance of the semi-conductive coating, the induced current as shown in Figure 16 is of such significance that the resulting sparks with the help of ozone can lead to accelerated degradation of the coating. This, in turn, will induce slot discharge and the affected machine can eventually fail only after a few years of operation.

Since the physics of the vibration sparking is not voltage driven, such contact sparking can occur at the low voltage end of the winding and the resulting PD pattern may not be locked to the AC sine wave. For this reason, early stage detection is complicated though eventually slot discharge should be reliably recognized as in 4.2.2.2.

Failure mechanism	Polarity	Load Effect	Temperature Effect	Phase Location
Vibration Sparking	Unknown	Direct	Inverse	Unknown

4.5 Extraneous Signals

As described in Section 3.5, “Disturbances are electrical pulses of relatively short duration that may have many of the characteristics of stator winding PD pulses – but in fact are not stator winding PD” [12]. Some of these disturbances are synchronized to the AC cycle, and some are not. Sometimes synchronized disturbance pulses can be suppressed based on their position with respect to the AC cycle. See Section 5.1.2 for more information about Disturbance separation.

Hence, as part of the data analysis process, it is important to identify whether the pulse patterns displayed contain disturbances from sources within the stator winding environment that could not be separated by the PD instrument or test configuration. Disturbance-like sources tend to produce patterns that occur across the AC cycle. There are three common disturbance patterns that may occur.

4.5.1 Excitation pulses

PD couplers sometimes detect the high frequency thyristor-firings of static excitation systems. These pulses are usually of equal pulse repetitions, low magnitudes and occur repetitively across the AC cycle (Figure 17). Usually, they are only discernible when there is low PD activity on the unit and the test range of the PD test instrument is set to its most sensitive (lowest range) setting. This is not usually indicative of a problem with a stator winding, but if pulse levels are high this may indicate that there is a problem in the exciter. Although this type of pattern is called “excitation pulses”, similar signals may also be detected from machines close to variable frequency drives in the electrical system.

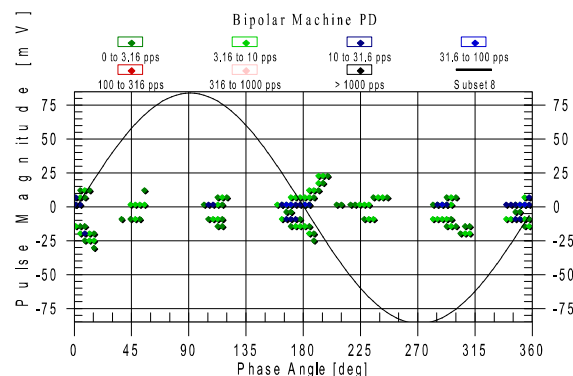


Figure 17: Excitation Pulses

Disturbance pulses are not indicative of PD sources within the winding insulation.

4.5.2 Cross-coupling

PD pulses are high frequency pulses that may capacitively couple from one phase to another across the endwinding or circuit rings. In addition, a coupler connected to one phase may detect pulses radiated from a source on another phase. This behaviour is called cross-coupling and can be detected by observing PD pulses shifted $\pm 120^\circ$ from the positions of the source PD. Phase-to-ground PD cross-coupled from another phase would cluster at 345° , 105° , 165° or 285° . Cross-coupled inter-phase activity would be $\pm 30^\circ$ phase-shifted from these positions as shown for C-phase (middle plot at 315°) in Figure 28.